

Searching Every Seed at a Fixed Length: A Guide to Paradoxical Cylinders

Collatz proof project*

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1 What we are looking for

Sometimes the powers of three and two suggest a Collatz segment should shrink, yet its endpoint is at least its starting value. The added ones make up the difference. Following Rozier and Terracol, we call this a paradoxical segment. Earlier values are allowed to fall below the seed.

Their paper proves that if only finitely many such segments start above two, Collatz follows. That makes this a useful research target, but its finiteness remains unproved. See <https://arxiv.org/html/2502.00948v3>.

2 How a finite search can cover arbitrarily large seeds

Fix a length L . Seeds with the same remainder r on division by 2^L have the same first L parities. Write each as $r + 2^L q$. Their endpoints then have the form $u + Aq$, for known integers u and A .

When $A < 2^L$, the condition that the endpoint is at least the seed becomes

$$r + (2^L - A)q \leq u.$$

This gives a finite interval of possible q values. Checking all remainders and their intervals therefore covers every seed at this length, with no arbitrary seed-size ceiling. Lean proves the interval formula and the coverage identity.

3 What has been checked

Lean proves that the only seeds above two producing a paradoxical segment of length eight are

$$7, 9, 18, 19, 25.$$

Each of these had already descended before the endpoint. A separate tested Python search through length twenty finds the same five segments and no others. The full twenty-length computation is not a Lean proof; the length-eight classification is.

This reproduces a small known case and provides a checked search tool. The number of remainders doubles with every extra step, so simply searching longer will not settle finiteness across all lengths. The next page explains a proved pruning rule that makes longer searches practical. A restriction uniform in length is still missing.

*Prepared with Codex (OpenAI). Companion to the formal search note; no claim of a new mathematical discovery or a Collatz proof.

4 Skipping impossible completions

We can now reject many unfinished parity words without enumerating their endings. Suppose a prefix already has correction D , length s , and a odd steps. If the unfinished part has length h and k odd steps, its total correction cannot exceed

$$U_k = 3^k D + 2^{s+h-k} (3^k - 2^k).$$

Lean proves this bound for every possible arrangement of those ending letters. It is generous: the test gives the unfinished word the largest correction it could obtain.

Let m be the smallest seed at least three with the current prefix. When the whole word would have a contracting coefficient, a paradoxical endpoint needs correction at least

$$(2^{s+h} - 3^{a+k})m.$$

If even U_k is smaller, that choice of k cannot work. Lean now proves that only the largest allowed odd count needs testing: it permits the largest correction and requires the smallest correction to avoid descent.

The same inequality also bounds every possible seed in the unfinished class. When at most sixteen candidates remain, the search checks them directly and finishes that branch. Exact eight-step lookups accelerate these checks; every reported segment is replayed one step at a time.

5 What the improved search found

At length 27, the search finds the same 50 segments using 955 tree nodes and 5,159 direct candidate checks. At length 40, it finds none using 2,889 nodes and 15,703 candidate checks.

The previously incomplete length-65 census now finishes: 5,324,915 tree nodes and 27,386,515 candidate checks find 244 segments. Their seeds range from 73 to 4547. All 244 had already descended below their seed before the endpoint. No branches remain pending. These are results at selected lengths; intervening lengths were not all checked.

6 How much is proved

The correction bound and pruning implication are Lean theorems. The larger searches are Python computations checked against full enumeration at smaller lengths, with the affine identity checked at each visited node and every reported segment replayed directly. They are not kernel-verified census certificates.

This gives a better instrument for investigating long words. It does not prove that every sufficiently long word is excluded, so it does not finish the finiteness argument or prove Collatz. The next page explains a proved limitation of this test. The formal note records the exact statements and reproduction commands.

7 Why this test alone cannot finish the proof

We have now proved a limitation of the pruning rule. Fix any prefix and its smallest allowed seed. If the target length is taken sufficiently large, at least one choice of the number of odd steps in the unfinished part always passes the correction-envelope test.

The reason is that the envelope lets us arrange the unfinished odd steps in the most favorable possible way. Enough odd steps make its permitted correction large enough to offset the apparent contraction. Adding more allowed even steps makes the total coefficient contracting again, while the correction allowance is still large.

This does not construct a real orbit. The favorable arrangement may require a different seed residue, or fail other arithmetic constraints. It demonstrates that the envelope forgets information which an eventual proof will need.

8 The precise consequence

For any fixed finite collection of prefixes, there is a length beyond which all their relaxed tests survive. Therefore this test alone cannot exclude every sufficiently long word using one fixed finite prefix tree.

The statement does not say that all search methods fail, or that deeper pruning is useless. The successful fixed-length searches remain valid. It also says nothing about whether Collatz is true or false.

9 Where to go next

The next argument should connect a suffix's possible correction to the residue of the seed that actually realizes it. Alternatively, it needs another restriction valid uniformly in length. More computation can help study those constraints, but simply increasing the current search limit is not a proof of the missing uniform statement.

The limitation and its finite-family consequence are Lean theorems in `Collatz.PruningLimit`. They are included in the expanded 124-declaration audit. This result changes the research direction by identifying precisely what the present bound cannot establish on its own.